Task : Advanced SOC Operations and Threat Intelligence

1. Objective

The objective of this week's training is to develop advanced Security Operations Center (SOC) skills, including log analysis, threat intelligence integration, and incident escalation workflows. This report covers both theoretical understanding and practical implementation using industry-standard tools.
The practical section focuses on applying these concepts in a simulated environment to enhance detection, analysis, response, and reporting capabilities.

2. Advanced Log Analysis

2.1 Overview

Advanced log analysis involves collecting, correlating, and analyzing logs from multiple sources to identify complex attack patterns. It is a core function in SOC operations, enabling analysts to detect threats that may not be visible from a single log source.

2.2 Log Correlation

Log correlation is the process of combining logs from different systems such as firewalls, endpoints, and servers to identify related events.
For example, multiple failed login attempts (Event ID 4625) combined with unusual outbound traffic may indicate a brute-force attack followed by data exfiltration.

2.3 Anomaly Detection

Anomaly detection focuses on identifying unusual behavior that deviates from normal system activity. Examples include:

- Logins at unusual hours
- High-volume data transfers
- Unexpected process execution

These anomalies may indicate potential security incidents.

2.4 Log Enrichment

Log enrichment adds contextual information to raw logs, such as:

- Geolocation of IP addresses
- User roles and privileges
- Threat intelligence data

This improves the accuracy and efficiency of analysis.

2.5 Advanced Considerations

- Use of machine learning for anomaly detection
- Correlation rules in SIEM systems
- Reduction of alert noise through filtering

2.6 Key Objective

To develop the ability to analyze and correlate logs effectively for detecting complex threats.

3. Threat Intelligence Integration

3.1 Overview

Threat intelligence integration enhances SOC capabilities by providing contextual data about threats, including indicators of compromise (IOCs), attacker techniques, and known malicious infrastructure.

3.2 Threat Intelligence

- **IOCs:** IP addresses, file hashes, domains
- **TTPs:** Attacker techniques mapped using MITRE ATT&CK
- **Threat Feeds:** Real-time updates from platforms like AlienVault OTX

3.3 Integration in SOC

Threat intelligence is integrated into SIEM tools to:

- Enrich alerts
- Improve detection accuracy
- Identify known malicious activity

3.4 Threat Hunting

Threat hunting uses intelligence to proactively search for threats.
Example:

- Detect misuse of valid accounts (T1078)
- Process Injection (T1055)
- Data Compressed for Exfiltration (T1002)

3.5 Advanced Considerations

To improve analysis, SOC analysts also include:

- Automated threat feed ingestion
- Correlation with internal logs
- Intelligence-driven detection rules

3.6 Key Objective

To understand how threat intelligence improves detection and enables proactive defense.

4. Incident Escalation Workflows

4.1 Overview

Incident escalation ensures that security incidents are handled by the appropriate level of expertise within a SOC.

4.2 SOC Tiers

- **Tier 1:** Initial triage
- **Tier 2:** Investigation and analysis
- **Tier 3:** Advanced threat analysis

4.3 Escalation Criteria

- Severity of incident
- Complexity
- Potential impact

4.4 Communication Protocols

Structured communication includes:

- Situation Reports (SITREP)
- Incident summaries
- Stakeholder briefings

4.5 Automation in Escalation

Automation tools can:

- Assign tickets
- Enrich alerts
- Trigger workflows

4.6 Advanced Considerations

- Integration with SOAR platforms
- Automated escalation rules
- Real-time collaboration

4.7 Key Objective

To develop skills in managing escalation workflows and effective communication during incidents.

5. Advanced Log Analysis

5.1 Overview

Advanced log analysis involves collecting and analyzing logs from multiple sources to detect complex security threats. In this task, log analysis was performed using Elastic Cloud to simulate real-world SOC monitoring.

Figure 1: elastic cloud interface

5.2 Log Correlation

Logs were analyzed to identify failed login attempts using Event ID 4625. These logs were filtered and correlated based on timestamps and source IP addresses.

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-04-03 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious activity

Figure 2: Google sheet Table Format

This correlation helps in identifying potential brute-force attacks.

Figure 3: Filtering of different details of product dataset

5.3 Anomaly Detection

An anomaly detection rule was created to detect high-volume data transfers. The rule was configured to trigger when data transfer exceeded a defined threshold.
This helps identify unusual behavior that may indicate data exfiltration.

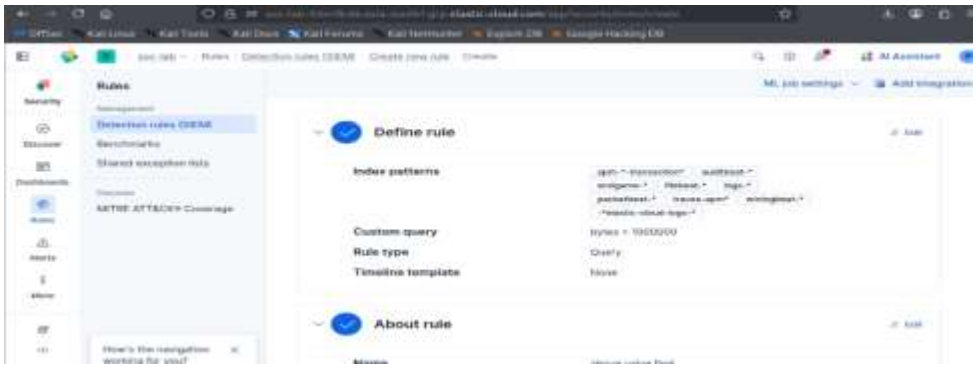

Figure 4: custom rule creation

5.4 Log Enrichment

Log enrichment was implemented using IPinfo to add location data to IP addresses. This provides additional context for analysis and improves detection accuracy.

Figure 5: Geo location IP data creation

5.5 Advanced Considerations

- Correlation across multiple log sources improves detection
- Anomaly-based detection identifies unknown threats
- Enrichment enhances decision-making

5.6 Summary

Log analysis identified failed login attempts and correlated them with suspicious activity. An anomaly detection rule was created to detect high data transfers. Log enrichment using GeoIP added context to IP addresses. These steps demonstrate how SIEM tools enhance threat detection through correlation, anomaly identification, and contextual analysis.

5.7 Key Objective

To develop practical skills in log analysis, correlation, anomaly detection, and enrichment using SIEM tools.

6. Threat Intelligence Integration

6.1 Overview

Threat intelligence integration enhances SOC operations by providing contextual information about potential threats. In this task, threat intelligence was integrated using AlienVault OTX and analyzed within Wazuh.

Figure 6: wazuh Dashboard

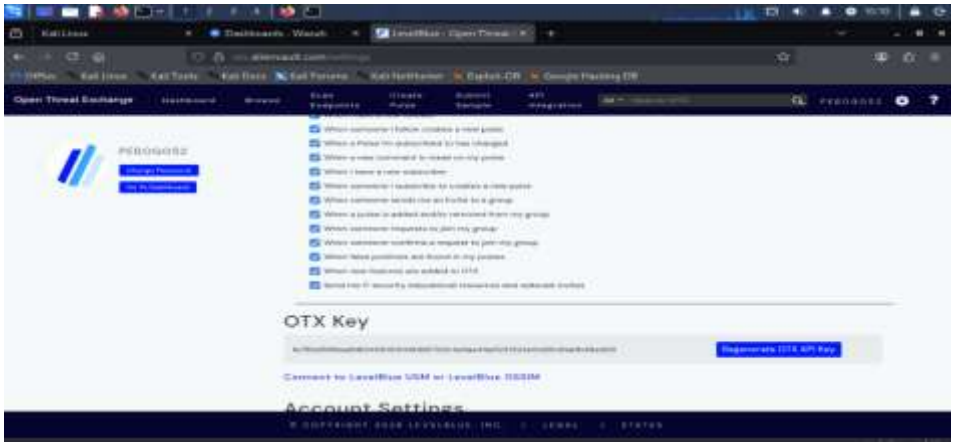

Figure 7: AlienVault OTX key genration

6.2 Threat Feed Analysis

A sample IP address was analyzed using AlienVault OTX to determine its reputation. The platform provided information about

Figure 8: IP address detail checking

6.3 Alert Enrichment

The alert was enriched by adding threat intelligence data:

Alert ID	IP	Reputation	Notes
003	192.162.99.230	Malicious (OTX)	Linked to suspicious activity

This enrichment improves the ability to prioritize and respond to alerts.

6.4 Threat Hunting

Threat hunting was performed by querying logs for abnormal user activity. The query identified potential misuse of valid accounts, which is associated with MITRE technique T1110

Figure 9: Querying abnormal activity in wazuh & log view

6.5 Advanced Considerations

- Automated threat feed integration improves efficiency
- Intelligence-based detection reduces false positives
- Continuous threat hunting strengthens security posture

6.6 Summary

Threat intelligence integration was performed using AlienVault OTX to validate suspicious IP activity. Alerts were enriched with reputation data, improving analysis accuracy. Threat hunting was conducted by identifying abnormal user activity. This demonstrates how integrating external intelligence enhances detection, investigation, and proactive threat identification in SOC environments.

6.7 Key Objective

To understand how threat intelligence enhances alert analysis, supports threat hunting, and improves SOC detection capabilities.

7. Incident Escalation Practice

7.1 Overview

Incident escalation ensures that security events are handled by the appropriate level of expertise. In this task, escalation workflows were simulated using TheHive.

Figure 10: TheHive DashBoard

7.2 Case Creation

A high-priority incident case was created to simulate unauthorized access. The case included relevant details such as severity, description, and associated MITRE technique.

Figure 11: Case Creation in TheHive

7.3 Escalation Process

The incident was escalated from Tier 1 to Tier 2 after initial triage. A structured escalation note was prepared to communicate key details and required actions.

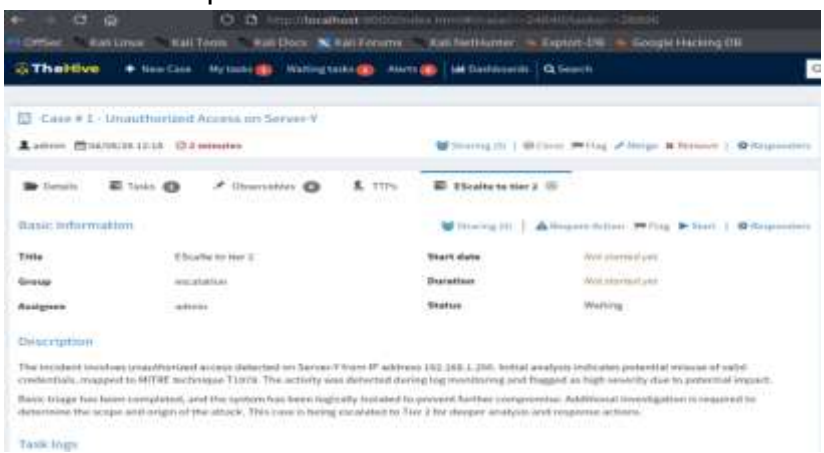

Figure 12: Case Escalation to Tier 2 With Escalation note added of 100 word

7.4 SITREP Documentation

A Situation Report (SITREP) was created using Google Docs. It included incident summary, actions taken, and current status.

Situation Report (SITREP)

Title: Unauthorized Access on Server-Y

Summary:

A high-priority incident involving unauthorized access was detected on Server-Y at 13:00. The activity originated from IP address 192.168.1.200 and is associated with potential misuse of valid credentials.

Details:

- Detection Time: 2025-08-18 13:00
- Source IP: 192.168.1.200
- MITRE Technique: T1078
- Severity: High

Actions Taken:

- System isolated
- Initial triage completed
- Incident escalated to Tier 2

Status:

Under investigation

Figure 13: SITREP Document

7.5 Automation Concept

Automation can be implemented using SOAR platforms to automatically assign high-priority alerts to Tier 2 analysts. This reduces response time and ensures consistent handling of critical incidents.

7.6 Advanced Considerations

- Structured escalation improves response efficiency
- Clear communication reduces delays
- Automation enhances consistency in SOC workflows

7.7 Key Objective

To understand and implement incident escalation workflows, including case creation, communication, and documentation.

8. Alert Triage with Threat Intelligence

8.1 Overview

Alert triage is the process of analyzing and prioritizing alerts to determine their severity and validity. In this task, triage was performed using Wazuh along with threat intelligence platforms.

8.2 Triage Simulation

A simulated alert involving suspicious command execution was analyzed. The alert was categorized based on its potential impact and assigned a high priority.

Alert ID	Description	Source IP	Priority	Status
808	Command Execution	192.168.40.139	Low	Open

Figure 14: Mock Alert

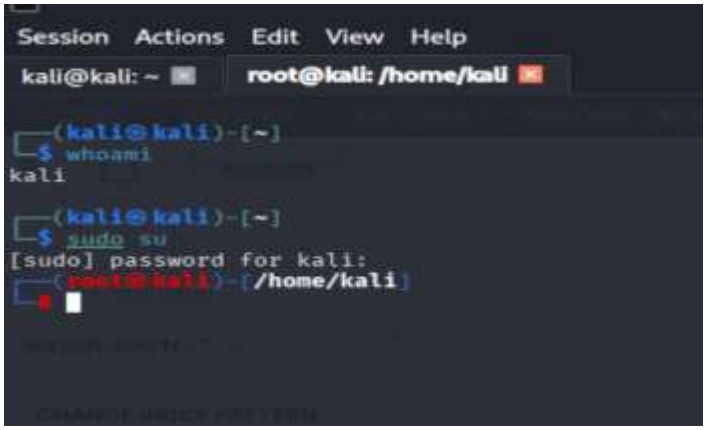

Figure 15: log creation

Figure 16: Log Alert in Wazuh

8.3 IOC Validation

The associated IP address was validated using VirusTotal and AlienVault OTX. These platforms provided reputation data to determine whether the IP was malicious.

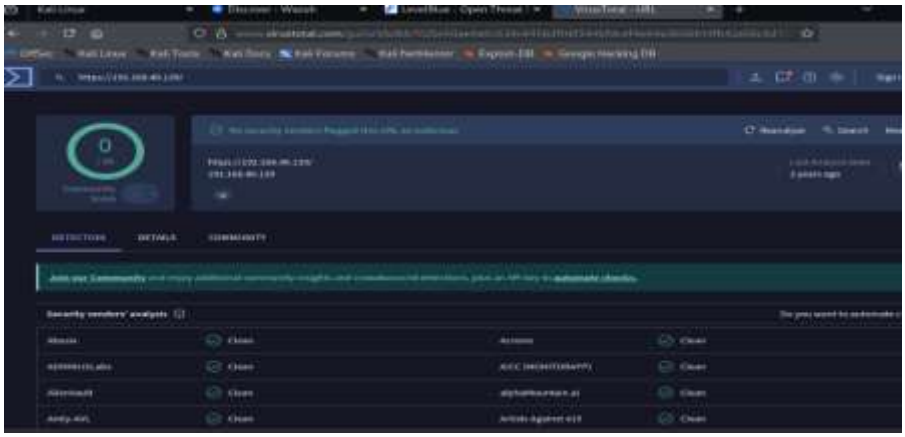

Figure 17: IOC validation with VirusTotal

8.4 Analysis Outcome

Based on the validation results, the alert was confirmed as suspicious and required further investigation. This demonstrates how threat intelligence supports accurate decision-making.

8.5 Advanced Considerations

- Threat intelligence reduces false positives
- Cross-platform validation improves accuracy
- Automated enrichment enhances SOC efficiency

8.6 Summary

Alert triage was performed by analyzing a simulated suspicious command execution event. The associated IP address was validated using VirusTotal and AlienVault OTX to determine its reputation. The findings helped classify the alert priority, demonstrating how threat intelligence improves investigation accuracy and reduces false positives.

8.7 Key Objective

To perform effective alert triage and validate indicators of compromise using threat intelligence platforms.

9. Evidence Preservation and Analysis

9.1 Overview

Evidence preservation ensures that collected data remains intact and reliable for investigation. In this task, evidence was collected and analyzed using Velociraptor

Figure 18: Veliciraptor dashboard

9.2 Volatile Data Collection

Volatile data was collected using Velociraptor by capturing active network connections. This data provides insights into system activity at the time of investigation.

Figure 19: Active network connection netstat download

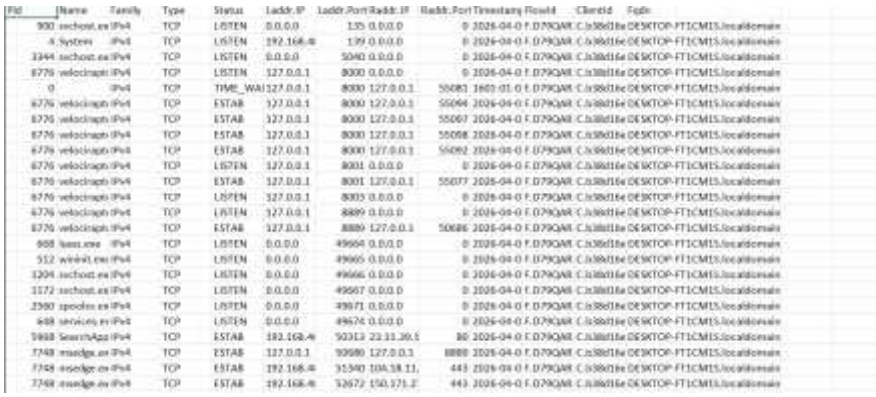

Figure 20: exported data of network netstat in csv

9.3 Memory Acquisition

A memory dump was collected to capture system state, including running processes and active sessions. This is critical for advanced forensic analysis.

Figure 21:Memory acquisition collecting

9.4 Data Integrity

The collected memory dump was hashed using SHA-256 to ensure integrity. Hashing verifies that the evidence has not been altered.

SHA256 Hash 407f97aee04350fd663c43663e7126bda0f2d795e047da1af18f65a271eaba65

Figure 22: Hash value of memory acquistion

9.5 Chain of Custody

A structured record was maintained documenting evidence details such as description, date, and hash value.

Evidence Preservation Record

Item	Description	Collected By	Date	Hash Value
Network Data	Netstat output	SOC Analyst	Today	f91b7a9d33eaffec2e55cdf2c8614cca6f21c4190ec42236acc664f0ef454c65
Memory Dump	Windows VM Memory	SOC Analyst	Today	407f97aee04350fd663c43663e7126bda0f2d795e047da1af18f65a271eaba65

Figure 23: Evidence Record

9.6 Advanced Considerations

- Proper evidence handling ensures forensic reliability
- Hashing guarantees data integrity
- Secure storage prevents unauthorized modification

9.7 Key Objective

Evidence preservation was performed by collecting volatile network data and a memory dump using Velociraptor. The collected data was securely stored and verified using SHA-256 hashing. Chain-of-custody documentation was maintained to ensure integrity. This process demonstrates proper forensic handling and reliable evidence collection in incident response.

9.8 Key Objective

To perform forensic evidence collection, preserve data integrity, and maintain proper documentation for investigation purposes.

10. Capstone Project: Full SOC Workflow Simulation

10.1 Overview

This project simulates a complete SOC workflow, including attack execution, detection, triage, response, escalation, and reporting. The objective is to demonstrate practical SOC operations using real tools and controlled environments.

10.2 Attack Simulation

A vulnerability in the system was exploited using Metasploit. The Samba usermap script exploit was used to gain unauthorized access.

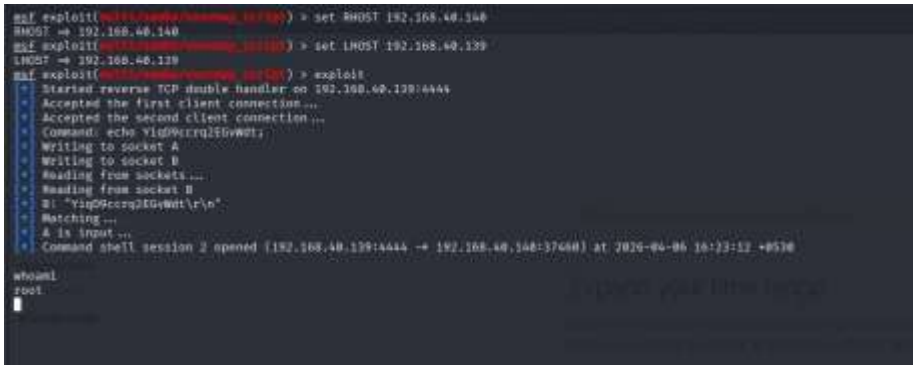

Figure 24: Samba Vulnerability exploitation successful

10.3 Detection and Triage

The attack was detected using Wazuh, and relevant logs were analyzed.

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	192.168.40.139	Samba exploit	T1210

10.4 Response and Containment

The affected system was isolated, and attacker activity was blocked to prevent further compromise.

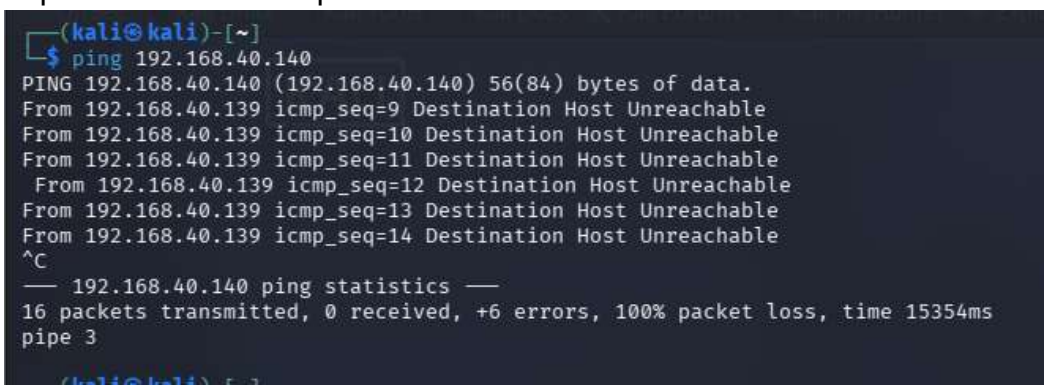

Figure 25: System Isolated

10.5 Escalation

The incident was escalated using TheHive for further investigation.

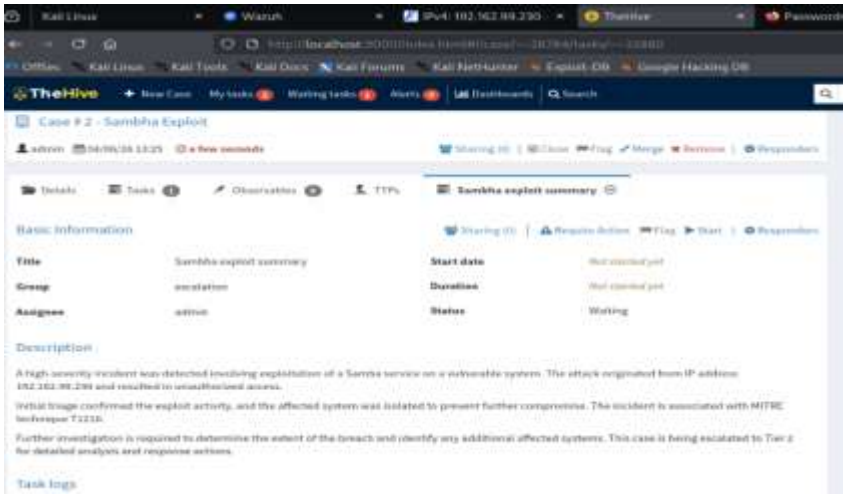

Figure 26: Escalation to Tier 2 with Summary

10.6 Reporting

A structured report was created using Google Docs to document the incident.

Incident Report – Samba Exploit

A simulated cyber attack was conducted to evaluate the effectiveness of SOC operations. The attack targeted a vulnerable Samba service on the Metasploitable system using a known exploit.

The attacker successfully gained unauthorized access, demonstrating how unpatched services can be exploited. The activity was detected through log monitoring and classified as a high-severity incident.

The affected system was isolated to prevent further compromise. Initial triage and analysis were performed to understand the nature and impact of the attack.

The incident highlights the importance of regular patching, monitoring, and rapid response. Although the attack was conducted in a controlled environment, it reflects real-world scenarios where attackers exploit known vulnerabilities.

This exercise demonstrates the complete SOC workflow, including detection, triage, response, escalation, and reporting.

Figure 27: Incident Report

STAKEHOLDER BRIEF

BRIEFING

A simulated cyber attack was carried out to test the organization's security response capabilities. The attack targeted a vulnerable service and resulted in unauthorized access to the system.

The issue was quickly identified and contained, ensuring no further impact. The system was isolated, and necessary actions were taken to prevent similar incidents.

This exercise highlights the importance of strong security practices, including regular updates and monitoring. Overall, the response process worked effectively, demonstrating the ability to detect and handle security incidents in a timely manner.

Figure 28: Briefing for a non-technical manager

10.7 Advanced Considerations

- Continuous monitoring prevents re-exploitation
- Threat intelligence improves detection accuracy
- Structured workflows enhance response efficiency

10.8 Key Objective

To simulate a complete SOC workflow and demonstrate the ability to handle security incidents from detection to reporting.

11. Challenges Faced

- Managing system resources while running tools like Wazuh and TheHive
- Configuring multiple virtual machines and ensuring proper network connectivity
- Initial setup and usage complexity of Velociraptor
- Understanding and interpreting threat intelligence data from AlienVault OTX
- Troubleshooting tool installation and execution issues

12. Key Learnings

- Gained hands-on experience with SIEM tools like Elastic Security and Wazuh for log analysis and correlation
- Learned to use threat intelligence platforms such as AlienVault OTX and VirusTotal for IOC validation
- Understood incident escalation workflows using TheHive
- Practiced evidence collection and integrity verification using Velociraptor
- Developed ability to perform end-to-end SOC workflow (detection → triage → response → reporting)

13. Conclusion

This report successfully demonstrates advanced SOC operations through practical implementation and theoretical understanding. Key areas such as log analysis, threat intelligence, escalation, and incident response were effectively covered.

The use of real tools and simulated environments provided a clear understanding of real-world SOC workflows. Overall, this training enhanced technical skills and prepared for practical cybersecurity roles.